

TaxiPulse: Real-Time Taxi Demand Forecasting with Fare and ETA Prediction

Zeroth Review

Group-6

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Introduction to the Problem Domain

Domain: Big Data Analytics in Urban Transportation (Smart Mobility)

- ▶ Urban transportation generates massive continuous data: every taxi / ride-hailing trip produces a record (pickup, drop, time, distance, fare).
- ▶ A single city like New York generates **100+ million trip records per year** — far beyond the capacity of traditional tools (Excel, pandas, RDBMS).
- ▶ Ride-hailing platforms (Uber, Ola, Lyft) depend on analytics over this data for driver allocation, upfront pricing, and surge management.

Why this domain matters

- ▶ Mismatch between taxi supply and demand $\Rightarrow$ idle drivers in quiet zones, long passenger waits in busy zones, lost revenue.
- ▶ Cities need demand insights for transport planning and policy.
- ▶ The domain naturally exhibits all 4 V's of Big Data — **Volume, Velocity, Variety, Veracity** — making it an ideal candidate for distributed analytics with Hadoop and Spark.

Problem Statement

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Taxi fleets and ride-hailing platforms cannot efficiently position drivers or quote reliable prices because taxi demand fluctuates sharply with time, location, and weather. Existing small-scale analyses cannot process city-scale trip data (20+ GB, hundreds of millions of records) or react to live conditions. There is a need for a **scalable big data pipeline** that processes massive historical trip data **and** a live trip stream to forecast **zone-wise taxi demand for the next hour** and predict the **fare and ETA** of individual trips in real time.

The 4 V's in our problem

- ▶ **Volume:** 20–50 GB of NYC trip records (200M+ trips, 2022–2024).
- ▶ **Velocity:** live trip stream processed with Spark Structured Streaming.
- ▶ **Variety:** structured trip records (Parquet/CSV) + semi-structured weather and zone data (JSON/GeoJSON).
- ▶ **Veracity:** noisy records — negative fares, zero-passenger trips, invalid GPS coordinates, missing values.

Existing System / Literature Review

Work	Contribution	Limitation / Gap
<i>Math. Biosciences & Eng.</i> (2024) — ST-enhanced GCN	Graph convolutional network with external spatio-temporal factors for ride-hailing demand prediction	Demand only; GPU-centric deep model; ignores fare/ETA; not real time
Guo et al., <i>Applied Soft Computing</i> (2025)	Multi-gated deep graph network with attention mechanisms for taxi demand prediction	Demand only; offline training/evaluation; no distributed big data pipeline
<i>arXiv:2407.01262</i> (2024) — Deep + tree fusion for ETA	Complementary fusion of deep networks and tree models for ETA prediction	ETA only; no demand or fare modelling; no streaming deployment
<i>Springer LNNS</i> (2025) — Trip duration model comparison	Compares Decision Tree, Random Forest, LightGBM and Gradient Boosting for NYC taxi trip duration prediction	Single-machine study; duration only; no live prediction or dashboard
<i>Int. J. Info. Eng. & Science</i> (2025)	Scalable fare prediction on NYC taxi trips using Spark and distributed ML	Fare only; no demand forecasting, no streaming, no visualization

Research Gaps Identified

- ▶ **Fragmented solutions:** prior works solve demand *or* fare *or* ETA in isolation — no unified system covers all three.
- ▶ **Scale gap:** most studies sample the data down to what fits on one machine instead of processing the full corpus distributively.
- ▶ **No real-time layer:** models are trained and evaluated offline; few works consume a live stream and update predictions continuously.
- ▶ **Weather ignored:** demand is highly weather-sensitive, yet many models exclude external context data.
- ▶ **No end-user visualization:** results stay in papers/notebooks; rarely delivered as a live decision-support dashboard.

Our opportunity

Combine all three predictions in one **distributed batch + streaming** pipeline on the full 20+ GB dataset, enriched with weather, delivered through a live dashboard.

Proposed System — TaxiPulse

Idea in one line: learn demand/fare/ETA patterns from 200M+ historical trips (batch), then apply them to a live trip stream to predict what happens *next* (real time).



Improvements over existing systems: full-scale distributed processing (no sampling); batch **and** streaming in one pipeline; weather-aware models.

What is new in TaxiPulse

- ▶ **Unified triple prediction:** one pipeline, one feature store, three outputs — zone-wise demand (next hour), trip fare, and ETA.
- ▶ **Hybrid batch + streaming design:** models trained on the full historical corpus are applied to a live Kafka stream, and live demand pace is compared against forecasts to raise **anomaly alerts** (e.g., event surge, transit outage).
- ▶ **Context-aware forecasting:** weather (rain/temperature) and calendar signals (holiday, weekend) fused with trip history.
- ▶ **Zone behaviour discovery:** K-Means clustering automatically labels zones as airport / office / nightlife types from their 24-hour demand signature.
- ▶ **Decision-support delivery:** live heatmap for drivers, fare–ETA widget for passengers, monitoring panel for operators.

Objectives of the Project

- ➊ **To design** a scalable distributed data pipeline that stores and processes 20+ GB of NYC taxi trip records using Hadoop HDFS and Apache Spark.
- ➋ **To implement** data cleaning and feature engineering at scale, quantifying and removing noisy/invalid records (veracity handling).
- ➌ **To develop** Spark MLlib models for (a) zone-wise hourly demand forecasting, (b) trip fare prediction, and (c) ETA prediction, comparing at least two algorithms per task.
- ➍ **To implement** a real-time layer using Kafka and Spark Structured Streaming that computes live zone statistics and applies the trained models.
- ➎ **To evaluate** model accuracy (RMSE/MAE) and pipeline performance, demonstrating tuning gains from partitioning, caching, and file formats.
- ➏ **To build** an interactive dashboard with a live demand heatmap and a fare/ETA prediction widget.

Big Data Tools & Purpose

Tool	Purpose
Hadoop HDFS	Distributed storage of raw trip data
Hadoop MapReduce	Baseline batch aggregation job
Apache Spark	Distributed cleaning, transformation, SQL analytics
Spark MLlib	Demand / fare / ETA models, K-Means clustering
Spark Structured Streaming	Real-time windowed processing
Apache Kafka	Live trip stream (replay) ingestion
Streamlit	+ Dashboard and heatmap visualization
Plotly/Folium	

Programming Languages

- ▶ Scala — core Spark data pipeline
- ▶ Python (PySpark) — ML and streaming

Dataset

- ▶ NYC TLC Trip Record Data (Yellow, Green, FHV; 2022–2024), $\approx$ 20–50 GB, 200M+ records
- ▶ NYC Taxi Zone lookup + GeoJSON
- ▶ Open-Meteo weather API (JSON)

Hardware

- ▶ 8+ GB RAM laptops (pseudo-distributed) / 3-node cluster

Project Timeline — Milestones

Week	Milestone	Deliverable
1–2	Environment setup (Hadoop, Spark, Kafka); dataset download and load into HDFS	Working cluster + data in HDFS
3–4	Data cleaning and feature engineering in Spark (Scala); data quality report	Cleaned feature tables
5	Spark MLlib models: demand, fare, ETA; algorithm comparison	Trained models + metrics
6	Kafka replay + Spark Structured Streaming; live predictions	Real-time pipeline
7	Performance tuning (partitioning, caching); dashboard build	Benchmarks + dashboard
8	Integration testing, documentation, final report and demo	Final report + demo

Reviews: Zeroth (now) → First (after Week 4)

Thank You

Questions & Suggestions

TaxiPulse — sensing the city's heartbeat, one trip at a time.